

DAA (LAB)

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EXPERIMENT:- DA2

DATE:- 13-MAR-2026

MAX_SUBARRAY:-

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SCOPE / BCSE204P_VL2025260504984 / Divide and conquer / Max_subarray

Max_subarray

DescriptionSubmission view

Grade

Reviewed on Tuesday, 3 February 2026, 10:52 AM by Automatic grade
Grade: 10.00 / 10.00

Assessment report [-]
[+]Summary of tests

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max_subarray.c

```
1 | #include <stdio.h>
2 | #include <limits.h>
3 | int maxCrossingSum(int arr[], int low, int mid, int high){
4 |     int leftSum=INT_MIN, rightSum=INT_MIN, sum=0;
5 |     for(int i=mid; i>=low; i--){
6 |         sum+=arr[i];
7 |         if(sum>leftSum){
8 |             leftSum=sum;
9 |         }
10 |     }
11 |     sum=0;
12 |     for(int i=mid+1; i<=high; i++){
13 |         sum+=arr[i];
14 |         if (sum>rightSum){
15 |             rightSum=sum;
16 |         }
17 |     }
18 |     return leftSum+rightSum;
19 | }
20 | int maxSubarraySum(int arr[], int low, int high){
21 |     if(low==high){
22 |         return arr[low];
23 |     }
24 |     int mid=(low+high)/2;
25 |     int leftMax=maxSubarraySum(arr, low, mid);
26 |     int rightMax=maxSubarraySum(arr, mid+1, high);
27 |     int crossMax=maxCrossingSum(arr, low, mid, high);
28 |     return (leftMax>rightMax ? (leftMax>crossMax ? leftMax:crossMax):(rightMax>crossMax ? rightMax:crossMax);
29 | }
30 | int main(){
31 |     int n;
32 |     scanf("%d", &n);
33 |     int arr[n];
34 |     for(int i=0; i<n; i++){
```

KARATSUBA ALGORITHM:-



Karatsuba Algorithm

Description

Submission view

Grade

Reviewed on Tuesday, 3 February 2026, 11:15 AM by Automatic grade
Grade: 100.00 / 100.00

Assessment report [-]

[-]Summary of tests

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karatsuba.c

```
1 #include <stdio.h>
2 #include <math.h>
3 #include <string.h>
4 int getlength(int num){
5     int lenght =0;
6     while(num !=0){
7         num/=10;
8         lenght ++;
9     }
10    return lenght;
11 }
12 long long karatsuba(int x, int y){
13     if(x<10 || y<10){
14         return x*y;
15     }
16     int n=fmax(getlength(x),getlength(y));
17     int half=n/2;
18     int xh =x/pow(10, half);
19     int xl = x%(int)pow(10, half);
20     int yh =y/pow(10, half);
21     int yl = y%(int)pow(10, half);
22     long long z0=karatsuba(xl,yl);
23     long long z1=karatsuba(xh,xh-yl-yh);
24     long long z2=karatsuba(xh,yh);
25     return z2*pow(10, 2*half)+(z1-z2-z0)*pow(10, half)+z0;
26 }
27 int main(){
28     int x,y;
29     scanf("%d",&x);
30     scanf("%d",&y);
31     long long result = karatsuba(x,y);
32     printf("%lld\n",result);
33 }
```



MATRIX CHAIN MULTIPLICATION:-



Matrix chain computation

Description

Submission view

Grade

Reviewed on Wednesday, 4 February 2026, 8:15 PM by Automatic grade

Grade: 100.00 / 100.00

Assessment report [-]

[+]Summary of tests

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matrix.c

```
1 #include <stdio.h>
2 #include <limits.h>
3 int min(int a, int b) {
4     return (a < b) ? a : b;
5 }
6 int matrixChainMultiplication(int p[], int n) {
7     int dp[n][n];
8     for (int i = 1; i < n; i++)
9         dp[i][i] = 0;
10    for (int L = 2; L < n; L++) {
11        for (int i = 1; i < n - L + 1; i++) {
12            int j = i + L - 1;
13            dp[i][j] = INT_MAX;
14            for (int k = i; k < j; k++) {
15                int cost = dp[i][k] + dp[k + 1][j]
16                    + p[i - 1] * p[k] * p[j];
17                dp[i][j] = min(dp[i][j], cost);
18            }
19        }
20    }
21    return dp[1][n - 1];
22 }
23 int main() {
24     int n;
25     scanf("%d", &n);
26     int p[n];
27     for (int i = 0; i < n; i++)
28         scanf("%d", &p[i]);
29     printf("%d\n", matrixChainMultiplication(p, n));
30     return 0;
31 }
32
```

VPL

